

Mark schemes

Q1.

(a) $\tan^{-1}\left(-\frac{1}{3}\right) = -0.32$

Sight of ± 0.32 or 18.43

M1

$x = 2.82, 5.96$

a correct answer AWRT

A1

–1 for any extra in range, ignore extra answers not in range.

[SC 161.57, 341.57 AWRT M1A1 (max 2/3)]

A1

3

(b) $3(\tan^2 x + 1) = 5 \tan x + 5$

$3 \tan^2 x - 5 \tan x - 2 = 0$

AG

B1

1

(c) $(3 \tan x + 1)(\tan x - 2) = 0$

Attempt at factorisation/formula

M1

$\tan x = 2, -\frac{1}{3}$

A1

$x = 1.11, 4.25, 2.82, 5.96$ AWRT

3 correct [SC $x = 1.11, 4.25$ + their two answers from (a)]

B1

*4 correct, no extras in range
[SC 161.57, 341.57, 63.43, 243.43
AWRT B1 (max 3/4)]*

B1

4

[8]

Q2.

(a) Area of sector = $\frac{1}{2}r^2\theta = \frac{1}{2} \times 6^2 \times \theta$
 $\frac{1}{2}r^2\theta$ *seen or used*

M1

$6 \times 3 = 2 \times \frac{1}{2} \times 6^2 \times \theta$
OE Forming equation

m1

$36\theta = 18 \Rightarrow \theta = 0.5$
AG

A1

3

(b) Arc = 6θ ;
 $r\theta$ seen or used

M1

..... = 3 cm
PI by a correct perimeter

A1

$\Rightarrow$ Perimeter = $12 + \text{arc} = 15$ cm

Ft wrong evaluation of 6θ . Condone missing/wrong units throughout the question.

A1F

3

[6]

Q3.

(a) $\frac{BC}{\sin 72} = \frac{18.7}{\sin 50}$ [= 24.4...]
Use of the sine rule

M1

$BC = \frac{18.7 \sin 72}{\sin 50}$
Rearrangement

m1

$$(BC) = 23.21(6..) \{= 23.2 \text{ to nearest } 0.1 \text{ cm}\}$$

AG Need > 1 dp if using cm eg 23.21 or 23.22; at least 1 dp if using mm.

A1

3

(b) Angle $C = 180^\circ - (50^\circ + 72^\circ) = 58^\circ$

Valid method to find either angle C (Pl eg by $\sin C = 0.848(04..)$ or side AB

M1

$$\text{Area of triangle} = 0.5 \times 18.7 \times 23.2.. \times \sin C$$

OE eg $0.5 \times 18.7 \times AB \times \sin 72^\circ$

M1

$$..... = 184 \text{ cm}^2$$

*Accept 183.8 to 184.2
Condone missing/wrong units*

A1

3

[6]

Q4.

(a) $\cot x = 2 \Rightarrow \tan x = 0.5$

M1

$$x = 0.46, 3.61$$

AWRT; no others within range

A1

2

(b) $\operatorname{cosec}^2 x = \frac{3 \cos x + 4}{2}$

$$2(1 + \cot^2 x) = 3 \cot x + 4$$

Correct use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$$(2 \cot^2 x - 3 \cot x + 2 - 4 = 0)$$

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

*AG; correct with no slips from line
with no fractions*

A1

2

- (c) $(2 \cot x + 1)(\cot x - 2) (= 0)$
Attempt to solve

M1

$$\cot x = -\frac{1}{2}, 2$$

A1

$$\tan x = -2, 0.5$$

$$x = 0.46, 3.61, 2.03, 5.18$$

2 correct Allow 3.6(0)

B1

*4 correct (with no extras in range) AWRT
 SC Degrees*

$$\left. \begin{array}{l} 26.57, 206.57 \\ 116.57, 296.57 \end{array} \right\} \text{B1 For 2 correct}$$

B1

4

[8]

Q5.

- (a) Arc $PQ = r\theta$
 $r\theta$

M1

$$= 6\pi \text{ (cm)}$$

Condone missing units throughout the paper

A1

2

- (b) $\alpha + \alpha + \frac{3\pi}{7} = \pi$
OE

M1

$$\alpha = \frac{2\pi}{7}$$

Accept equivalent fractions eg $\frac{4\pi}{14}$ and condone 0.286 π or better

A1

2

(c) Chord $PQ = 2 \times 14 \times \cos \alpha$

OE eg $2 \times 14 \times \sin \frac{3\pi}{14}$ or 17.45 – 17.5

inclusive or $\sqrt{14^2 + 14^2 - 2 \times 14^2 \cos \frac{3\pi}{7}}$

M1

Perimeter = 17.45... + 6 π
 = 36.307... = 36.3 (cm)
 Condone > 3sf

A1

2

[6]

Q6.

(a) $\{BC^2 =\} 7.6^2 + 8.3^2 - 2 \times 7.6 \times 8.3 \cos 65$
RHS of cosine rule used

M1

..... = 57.76 + 68.89 – 53.3175...
Correct order of evaluation

m1

$BC = \sqrt{73.33...} = 8.563.. (= 8.56 \text{ m})$

AG; must see $\sqrt{73.33...}$ or > 3sf value

A1

3

(b) Area triangle = $\frac{1}{2} \times 7.6 \times 8.3 \times \sin 65$

Use of $\frac{1}{2} bc \sin A$ OE

M1

$$= 28.58... = 28.6 \text{ (m}^2\text{)}$$

Condone > 3sf

A1

2

- (c) Area of triangle = $0.5 \times BC \times AD$
Or valid method to find $\sin B$ or $\sin C$

M1

$$AD = [\text{Ans (b)}] \div [0.5 \times \text{Ans (a)}]$$

Or $AD = 7.6 \sin B$; Or $AD = 8.3 \sin C$

m1

$$AD = 6.67... = 6.7 \text{ (m)}$$

If not 6.7 accept 6.65 to 6.69 inclusive.

A1

3

[8]

Q7.

(a) $x = \cos^{-1} \frac{1}{3}$
PI

M1

$$= 1.23, 5.05 \text{ (} 0.39\pi, 1.61\pi \text{)}$$

AWRT (−1 for each error in range)
SC 70.53, 289.47 B1

A1,A1

3

(b) $\sec^2 x - 1 = 2 \sec x + 2$
use of $\sec^2 x = 1 + \tan^2 x$

M1

$$\sec^2 x - 2 \sec x - 3 = 0$$

AG; CSO

A1

2

(c) $\sec^2 x - 2 \sec x - 3 = 0$
 $(\sec x - 3)(\sec x + 1) = 0$

attempt to solve

M1

$$\cos x = \frac{1}{3} \text{ or } -1 \quad \text{o.e}$$

A1

$$x = 1.23, 5.05,$$

(2 answers in range from (a)) AWRT

B1F

$$3.14 \quad (\pi)$$

*all correct and no extras in range
SC 70.53, 289.47, 180 B1*

B1

4

[9]

Q8.

(a) {Area of sector = } $\frac{1}{2} r^2 \theta$

M1

$$= 0.5 \times 36 \times 1.2 = 21.6 \text{ cm}^2$$

Condone missing/wrong units throughout the paper

A1

2

(b) Arc = $r\theta$

M1

$$= 6 \times 1.2 = 7.2$$

A1

$$\text{Perimeter} = 12 + 7.2 = 19.2 \text{ cm}$$

Ft on incorrect evaluation of 6×1.2

A1ft

3

[5]

Q9.

(a) $6^2 = 4^2 + 5^2 - 2(4)(5) \cos \theta$

Use of the cosine rule

M1

$$\cos \theta = \frac{4^2 + 5^2 - 6^2}{2(4)(5)}$$

Rearrangement

m1

$$\cos \theta = \frac{5}{40} = \frac{1}{8}$$

CSO AG (be convinced)

A1

3

(b) $\cos^2 \theta + \sin^2 \theta = 1$

Stated or used (PI)

M1

$$\sin^2 \theta = \frac{63}{64}$$

Or better

A1

$$\sin \theta = \frac{\sqrt{63}}{8} = \frac{\sqrt{9 \times 7}}{8} = \frac{3\sqrt{7}}{8}$$

AG (be convinced)

A1

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3

(c) Area of triangle = $0.5 \times 4 \times 5 \times \sin \theta$.

M1

$$= \frac{30\sqrt{7}}{8} \text{ cm}^2$$

.....

OE (Condone 9.92)

A1

2

[8]

Q10.

(a) $\{x =\} \cos^{-1}(0.3) = 1.266 \dots \{= \beta\}$

$\cos^{-1} (0.3)$ PI by eg 72° or 73°

M1

$$\{x\} = 2\pi - \beta$$

Condone degrees or mix.

m1

$$x = 1.27, 5.02$$

Accept 1.26 to 1.27 with 5.01 to 5.02 inclusive

A1

3

(b) (i) $m(\pi, -1)$

B1 for each coordinate

B1;B1

2

(ii) $\{x_Q\} = 2\pi - \alpha$

OE (unsimplified)

B1

1

(c) Stretch (I) in x -direction (II) scale

Need(I) & one of (II), (III)

M1

factor $\frac{1}{2}$ (III)

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A1

2

(d) $\cos 2x = \cos \frac{4\pi}{5} \Rightarrow 2x = \frac{4\pi}{5}$

OE. (From correct work)

B1

$$\Rightarrow x = \frac{2\pi}{5} (= \alpha)$$

Condone decimals/degrees

$$x = \pi - \alpha; \text{ OE}$$

$$\text{OE eg } 2x = 2\pi - \frac{4\pi}{5}$$

Correct quadrant;
condone degrees/decimals/mix

M1

$$x = \pi + \alpha; \quad x = 2\pi - \alpha; \text{ OE}$$

Need both (OE for $2x =$) with no extras
(quadrants) within the given interval.
Condone degrees/decimals/mix

m1

$$x = \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}$$

Need all 4 solutions for x but condone
unsimplified provided in terms of π
Ignore extra values outside the given interval.

A1

4

[12]

Q11.

(a) (i) $2 (\operatorname{cosec}^2 x - 1) + 5 \operatorname{cosec} x = 10$

M1

$$2 \operatorname{cosec}^2 x - 2 + 5 \operatorname{cosec} x - 10 = 0$$

$$2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$$

AG

A1

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2

(ii) $(2 \operatorname{cosec} x - 3)(\operatorname{cosec} x + 4) = 0$

Attempt to solve

M1

$$\operatorname{cosec} x = \frac{3}{2} \text{ or } -4$$

Condone answers with no method shown

A1

$$\sin x = \frac{2}{3} \text{ or } -\frac{1}{4}$$

AG

A1

3

- (b) $(\theta - 0.1) = 0.73, 2.41, -0.25, -2.89$ AWRT
2 correct values, may be implied later
(41.8, 138.2, -165.5, -14.5)

B1

$\theta = 0.83, 2.51, -0.15, -2.79$ AWRT
2 correct answers

B1

+ 2 correct answers and no extra within range

B1

3

[8]

Q12.

(a) Arc = $r\theta$

For $r\theta$ or 20θ or PI by 20×1.4

M1

$28 = 20\theta \Rightarrow \theta = 1.4$
 AG

A1

2

(b) Area of sector = $\frac{1}{2}r^2\theta$

$\frac{1}{2}r^2\theta$
 OE seen

M1

$= \frac{1}{2} 20^2 (1.4) = 280$
 (cm²)

Condone absent cm²

A1

2

(c) (i) Area triangle = $\frac{1}{2} \times 15 \times 20 \times \sin 1.4$

Use of $\frac{1}{2}ab \sin C$ OE

M1

$$(= 147.8....)$$

Shaded area = Area of sector – area of triangle

M1

$$= 280 - 147.8 = 132 \text{ (cm}^2\text{) (3sf)}$$

*Ft on [ans (b) – 147.8...] to 3sf
provided [...] > 0*

A1ft

3

$$(ii) \quad \{BD^2\} 15^2 + 20^2 - 2 \times 15 \times 20 \cos 1.4$$

RHS of cosine rule used

M1

$$= 225 + 400 - 101.98$$

Correct order of evaluation

m1

$$\Rightarrow BD = \sqrt{523.019....} = 22.86... = 22.9 \text{ (cm) to 3 sf}$$

Condone absent cm

A1

3

[10]

Q13.

[Note: Calc. set in wrong mode, penalise only once on the paper.]
Condone missing units throughout the question]

$$(a) \quad \text{Area of triangle} = \frac{1}{2} (12)(8)\sin\theta$$

Use of $\frac{1}{2} ab \sin C$ or full equivalent

M1

$$\sin\theta = \frac{20}{48} \quad [=0.41(666...)]$$

OE (giving 0.412 to 0.42)

A1

$$\Rightarrow \theta = 0.4297(7...) = 0.430 \text{ to 3sf}$$

AG(need to see >3sf value)

A1

3

(b) $\{AB^2 = 8^2 + 12^2 - 2 \times 8 \times 12 \times \cos \theta$

M1

$$= 64 + 144 - 174.5\dots$$

*Accept 33 to 34 inclusive if three values
not separate*

m1

$$\Rightarrow AB = 5.78\dots = 5.8 \text{ cm to 2sf}$$

*If not 2sf condone 5.78 to 5.79 inclusive.
Condone $\pm$*

A1

3

(c) (i) $\text{Arc } AD = 8\theta$

M1

$$= 3.44\dots = 3.4 \text{ cm to 2sf}$$

If not 2sf condone 3.438 to 3.44 inclusive

A1

2

(ii) $\text{Area of sector} = \frac{1}{2} r^2 \theta$

Stated or used [or 13.7(6..) seen]

M1

Shaded area = Area of triangle – sector area

Difference of areas

M1

$$\text{Shaded area} = 20 - 0.5 \times 8^2 \times \theta$$

$$= 6.2 \text{ cm}^2 \text{ to 2sf}$$

Condone 6.24 to 6.2472

A1

3

[11]

Q14.

(a) Area of sector = $\frac{1}{2}r^2\theta = \frac{1}{2} \times 5^2 \times \theta$
 $\frac{1}{2} r^2\theta$ seen or used

M1

$12.5\theta = 8.1 \Rightarrow \theta = 0.648$

AG Condone $\theta = 0.648$ used to show that area = 8.1

A1

2

(b) Arc = 5θ ;
 5θ

M1

..... = 3.24 cm

PI by a correct perimeter

A1

$\Rightarrow$ Perimeter = 10 + arc = 13.24 cm

CSO Condone missing/wrong units; condone 3sf i.e. 13.2 if no obvious error NMS 3/3

A1 ✓

3

[5]

Q15.

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(a) $\frac{\sin B}{4.8} = \frac{\sin 100}{12}$

Use of the sine rule

M1

$\sin B = \frac{4.8 \sin 100}{12} [= 0.39(392...)]$
 Rearrangement

m1

(angle ABC) = 23.19(8...) {= 23.2°.}

AG Need >1dp e.g. 23.19 or 23.20

A1

3

(b) Angle $C = 80^\circ - 23.2^\circ = 56.8^\circ$

*Valid method to find a relevant angle e.g. C
(Pl e.g. by correct sin C) or $23.2^\circ + 10^\circ$*

M1

Area of triangle = $0.5 \times 12 \times 4.8 \times \sin C$

OE e.g. $0.5 \times 4.8 \times 12 \times \cos (B + 10)$

M1

..... = $24.09 \dots = 24.1 \text{ cm}^2$. (to 3sf)

Condone missing / wrong units

A1

3

[6]

Q16.

(a) $32^2 = 24^2 + 24^2 - 2 \times 24 \times 24 \cos \theta$

M1

or $\sin \frac{1}{2} \theta = \frac{\frac{1}{2}(32)}{24}$

$\cos \theta = \frac{24^2 + 24^2 - 32^2}{2 \times 24 \times 24}$

$\dots\dots\dots = \frac{128}{1152} \{ = \frac{1}{9} \} (= 0.11)$

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m1

or $\frac{1}{2} \theta = \sin^{-1} \left(\frac{2}{9} \right) (= 0.7297 \dots)$

$\theta = 1.459 \dots = 1.46$ to 3sf

CSO AG (be convinced)

A1

3

(b) Arc = $r\theta$

M1

= $24 \times 1.459 \dots = 35 \text{ cm}$

Condone absent cm; 35 to 35.04

A1

2

(c) (i) Area of sector = $\frac{1}{2}r^2\theta$
Seen

M1

$$= \frac{1}{2}(24)^2(1.459\dots) = 420.3 = 420 \text{ cm}^2$$

Condone absent cm²; 420 to 420.48

A1

2

(ii) Area of triangle = $\frac{1}{2}(24)(24)\sin\theta$
OE

M1

[= 286. (...)]

Shaded area = area of sector – area of triangle

Dep on at least one of the previous two M marks. PI

m1

$$\left[= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta \right] = 134 \text{ cm}^2$$

Condone absent cm²

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3

[10]

Q17.

(a) Area = $\frac{1}{2} \times 5 \times 4.8 \times \sin 30^\circ$

Use of $\frac{1}{2}ab\sin C$ OE

M1

6 cm².

Condone absent cm².

[Note: Calculator set in wrong mode, penalise only once on the paper.]

A1

2

(b) $AB^2 = 5^2 + 4.8^2 - 2 \times 5 \times 4.8 \cos 30^\circ$

RHS of cosine rule used

M1

$$= 25 + 23.04 - 41.569$$

Correct order of evaluation

m1

$$= 6.4707 \dots$$

$$\Rightarrow AB = \sqrt{6.47 \dots} = 2.5437$$

$$= 2.54 \text{ cm to 3 sf}$$

Accept 'better' than 2.54

Condone absent cm

A1

3

[5]

Q18.

(a) $\text{Arc} = r\theta$

For $r\theta$ or 16θ or 16×1.5 OE multiplication

M1

$$1.5r + r + r (= 56)$$

For realising that perimeter is sum of two radii and arc.

M1

$$3.5r = 56 \Rightarrow r = 16$$

AG Completion (condone verification)

A1

3

(b) Area of sector = $\frac{1}{2} r^2 \theta$

$$\frac{1}{2} r^2 \theta \text{ OE seen}$$

M1

$$= \frac{1}{2} 16^2 (1.5) = 192 \text{ cm}^2.$$

Condone absent cm².

A1

2

[5]



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